Companion Workbook to "How to Do Statistical Evaluations in ECE/CS Papers"

Project-Type Tables, an Evaluation-Plan Worksheet, an Autopsy of a Fragile Evaluation, and Exercises

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Abstract

This is a companion workbook for the tutorial *How to Do Statistical Evaluations in ECE/CS Papers* (the "main paper"). It contains the student-facing scaffolding that did not fit in the main paper: a project-type translation table for matching common ECE/CS evaluation patterns to appropriate methods, an evaluation-plan worksheet to fill out before running experiments, an autopsy of a typical fragile evaluation and how to fix it, and a set of exercises that map to the main paper's sections. It is intended to be printed out, marked up, and used as a teaching artifact for student research projects. References to sections beginning with §. . . refer to the main paper.

1 Choose Your Evaluation Type

ECE/CS students work on a wide range of empirical projects, and the unit of analysis, the dominant metric, the most common mistake, and the recommended method differ noticeably across them. Table 1 lays out the most common project types side-by-side. Students should locate themselves in the table before reading the main paper, and reread the row that applies to their project before each milestone.

Translation boxes. Three short translations of the main paper's running scheduling example to other ECE/CS subfields.

Wireless networking. Unit of analysis: topology / seed. Primary metric: packet delivery ratio (PDR), latency, routing overhead, energy. Pitfall: treating packets as independent (§4.1). Good plot: PDR vs. mobility speed with paired bootstrap CIs.

ML / anomaly detection. Unit of analysis: held-out flow / device / user / time window. Primary metric: AP (under class imbalance), AUROC, recall at fixed false-positive rate. Pitfall: leakage across time or devices. Good plot: precision–recall curve with calibration plot.

Project type	Unit of analysis	Typical metrics	Common mistake	Recommended default
Wireless / network simulation	seed / topology / workload	delay, PDR, throughput, energy	treating packets as independent	paired seeds, bootstrap CIs, multiple regimes
Systems benchmark	run / machine / configuration	latency, throughput, p_{95}, p_{99}	ignoring warmup, order, thermal effects	randomized order, warmup discarded, CDFs, CIs
ML classifier	held-out instance / group / time window	accuracy, AP, F_1, AUROC, calibration	tuning the threshold on the test set	validation/test split, AP for imbalance, paired bootstrap
Scheduler / optimizer	workload instance / seed	makespan, latency, utilization	only testing easy workloads	factorial regimes, ablations, paired tests
Embedded / IoT experiment	device / run / environment	energy, packet loss, deadline misses	hidden hardware or environment confounders	blocking, calibration, repeated trials
Simulation paper	scenario / seed	depends on the model	trusting simulator output blindly	sanity checks, conservation, baseline reproduction

Table 1: Project-type translation table. Find the row that matches your project and use it as the lens through which to read the main paper. The "unit of analysis" column is what the main paper calls the unit of analysis (§4.1 of the main paper); the "recommended default" column is the minimum-viable method, not the only acceptable one.

Embedded systems. Unit of analysis: repeated run / device / environment block. Primary metric: energy, latency, missed deadlines. Pitfall: temperature, battery state, background tasks (main §**??**). Good plot: energy vs. latency tradeoff.

2 Anatomy of a Fragile Evaluation

A fragile evaluation often *looks* fine on first read. The headline sounds confident, the numbers are big, and the statistical machinery is either absent or perfunctory. Below is a typical example, lightly fictionalized, of the kind of single-line claim that recurs in beginning student papers. We then take it apart and rewrite it.

"Our protocol improves throughput by 35%."

The sentence has at least nine separate problems, none of which are visible on the surface:

1. One topology was used.
2. One seed was used.
3. The baseline is a strawman version of the closest published method.
4. No confidence interval is reported, on the difference or on either arm.
5. Only throughput is reported; delay, energy, fairness, and overhead are silent.
6. Parameters were tuned on the proposed method but not on the baseline.

7. Mobility, failure, or workload stress conditions were not tested.
8. The simulator was not validated against any analytical or published baseline.
9. Per-packet measurements were treated as independent; the effective n is one.

A defensible rewrite of the same claim, using the main paper's machinery, might read:

> *"Across 30 random seeds and 5 topologies (factorial design, $5 \times 30 = 150$ paired runs), our protocol increases mean throughput by 35% (95% bootstrap CI: 31–39%) over the strongest published baseline at moderate offered load $\rho = 0.6$. The improvement holds in 4 of 5 topologies; in the fifth (a sparse line topology) the protocols are within 1%. Routing overhead increases by 12% (CI: 9–15%); energy is within 2%. Tuning was performed on a held-out validation topology; the test topologies above were not used during tuning. Simulator delay and queue-length outputs match the analytical M/M/1 model within 1.5% in the no-contention limit."*

The rewrite is longer, but each clause closes a specific reviewer objection. The original sentence closed none.

3 Evaluation Plan Worksheet

Fill out this worksheet *before* running experiments. The intent is to catch ambiguity in the design phase, when changing the plan is cheap. The last item, in particular, separates a real hypothesis from a wish.

1. Main claim (one sentence, plain English):

2. Falsifiable hypothesis:

3. Primary metric:

4. Secondary metrics (including cost/overhead):

5. Unit of analysis (seed / trace / topology / user / ...):

6. Baselines (and why they are the strongest reasonable ones):

7. Independent variables / factors:

8. Regimes to test (load, distribution, scale, mobility, ...):

9. Number of seeds / runs and a one-line justification:

10. Pairing / blocking plan (CRN? same workloads? blocked by topology?):

11. Statistical summary (mean / median / p_{99} / AP / ...):

12. Statistical test, if any (paired Wilcoxon / Friedman / McNemar / bootstrap / none):

13. Figures to produce (CDF / interaction / Pareto / paired-difference / ...):

14. Ablations:

15. Stress tests, regimes where the method may fail:

16. Simulator / measurement validation steps:

17. Reproducibility artifacts (code, seeds, traces, hardware):

18. *What result would make us weaken or abandon the claim?*

The last item is the one that distinguishes a real empirical hypothesis from a wish, and it is the one students most often skip. If no experimental outcome would change the claim, the claim is not falsifiable and the experiment is decoration.

A filled-out example: FIFO vs. SRPT

For students who have not seen what level of specificity to aim for, the following is the worksheet completed for the running example used in the main paper.

1. Main claim: SRPT lowers mean job completion latency relative to FIFO under heavy-tailed workloads, with the gap growing in offered load.
2. Falsifiable hypothesis: at $\rho \geq 0.5$ under truncated-Pareto job sizes, mean per-seed latency under SRPT is lower than under FIFO.
3. Primary metric: per-seed mean job completion latency.
4. Secondary metrics: per-seed p_{99} latency; per-seed throughput; preemption count (overhead).
5. Unit of analysis: *seed*. Each seed redraws arrival times and job sizes; per-seed values are the independent observations for CIs and tests.
6. Baselines: FIFO (a reasonable floor) and a published shortest-job-first variant (the strongest reasonable point of comparison; not a strawman).

7. Independent variables: scheduling policy (FIFO, SRPT); offered load ρ; job-size distribution (truncated Pareto with $\alpha \in \{1.5, 2.5\}$, exponential).
8. Regimes: $\rho \in \{0.3, 0.5, 0.7, 0.9\}$ crossed with the three job-size distributions; full 4×3 grid.
9. Number of seeds: 30 per cell; chosen to give 80% power against a Cohen's $d = 0.5$ at $\alpha = 0.05$ (paired).
10. Pairing / blocking plan: common random numbers (one workload per seed, replayed under each policy); per-seed differences are the analysis unit.
11. Statistical summary: per-cell mean of paired differences (FIFO − SRPT) with 95% bootstrap CI on the difference; p_{99} reported per cell.
12. Statistical test: paired Wilcoxon signed-rank on per-seed latency differences; significance reported alongside, not in place of, the CI.
13. Figures: latency CCDF on log–log axes; interaction plot of mean latency vs. ρ with paired bootstrap CIs; bootstrap distribution of the per-seed difference.
14. Ablations: replace SRPT with shortest-job-first (no preemption) to isolate the contribution of preemption.
15. Stress tests: very high load ($\rho = 0.95$); exponential job sizes (where the SRPT advantage should shrink); very small job buffers.
16. Validation: simulator passes M/M/1 baseline reproduction in the exponential-size limit; conservation laws (jobs in = jobs completed + jobs in flight) hold to machine precision.
17. Reproducibility: `figures.py` regenerates every figure from a fixed seed range; full code, seeds, and dependency versions in the project repository.
18. What would weaken the claim? If the per-cell paired-difference CI overlaps zero at any $\rho \geq 0.5$ under truncated Pareto, or if the SRPT advantage at $\rho = 0.9$ is below 50%, we would weaken the strength of the claim and report it.

4 Exercises

These exercises map roughly to the structure of the main paper. They are short and meant to be done with paper, a calculator, and (where indicated) a Python REPL. Solutions are not provided; they are the students' work.

E1. Strengthen a weak claim (main §??). Rewrite the following sentence as a strong evaluation claim, using the main paper's reporting order:

> *"Our cache improves throughput by 25%."*

E2. Identify the unit of analysis (main §??). For each scenario below, name the unit of analysis and the size of n.

(a) One simulation run on one topology produces 10,000 packet delays.

(b) Twenty seeded simulation runs on one topology each produce 10,000 packet delays per run.

(c) Ten topologies, with five seeded runs each, with 10,000 packet delays per run.

(d) A learned anomaly detector trained on traffic from 100 devices and evaluated on traffic from 20 held-out devices, with thousands of flows per device.

E3. Plot choice (main §??). Pick the most informative plot for each: (a) latency comparison of two schedulers across orders of magnitude; (b) throughput vs. load for two protocols, with uncertainty; (c) classifier comparison on imbalanced data; (d) energy vs. latency for several configurations of a method.

E4. Paired or unpaired? (main §??).

(a) Two schedulers run on the same 30 seeded workloads.

(b) Two protocols run on different randomly drawn topologies.

(c) Two ML models evaluated on the same test set.

(d) Two algorithms evaluated on traces from disjoint pools of users.

For each, state which test you would use by default and why.

E5. Compute an effect size and a CI (main §??, Listing ??). You are given 20 paired per-seed measurements of FIFO and SRPT mean latency. Write the four-line numpy snippet that produces (i) the mean per-seed reduction, (ii) the relative reduction in percent, and (iii) a 95% percentile bootstrap CI on the mean reduction.

E6. Find the flaw.

> *"We ran one seed and report $p < 0.05$ over 10,000 packets."*

Identify every flaw in this sentence and propose a corrected version.

E7. Design a factorial experiment with a budget (main §??). Assume that one "run" means one policy executed on one workload, and that a paired seed therefore costs two runs (one per policy). You have a budget of 90 runs. Design a factorial experiment over (offered load $\in \{0.3, 0.6, 0.9\}$) $\times$ (job-size distribution $\in \{\text{exponential}, \text{truncated Pareto}\}$) for two policies, with paired seeds. How many complete paired seeds per regime can you afford, and how many runs are left over? Which interactions can you estimate from the resulting design?

E8. Reproducibility checklist (main §??). For a project of your choosing in your group, write a short reproducibility statement that lists exactly the information another student would need to re-run your experiments and obtain the same numbers. Compare it against the main paper's reproducibility section and Appendix **??** (the pre-submission checklist).

E9. Equivalence (main §??). A new lightweight model claims to match a heavy baseline. You measure mean test accuracy as 84.1% (baseline) vs. 83.7% (proposed), with 95% CI on the difference of $[-1.0\%, +0.2\%]$. The team agrees on an equivalence margin of 1.5 percentage points. Does the experiment support the equivalence claim? What if the margin were 0.5 percentage points instead?

E10. The autopsy, in your own area. Find a published claim in your subfield (or invent a plausible one) that has the same surface shape as the autopsy in §2 of this workbook. List the problems and write a defensible rewrite.